

Characterization of the Exoplanet 51 Pegasi b via Radial Velocity Analysis

Elliot Czarik
Student Number: s3966879

January 20, 2026

Abstract

We analyze radial velocity observations of the star 51 Pegasi to characterize its orbiting planet, 51 Pegasi b, the first exoplanet discovered around a main-sequence star in 1995. Using χ^2 minimization on the original dataset from Mayor & Queloz (1995), we determine an orbital period of $P = 4.2334 \pm 0.0010$ days and a radial velocity semi-amplitude of $K = 56.59^{+0.67}_{-0.67}$ m/s. Statistical testing via the χ^2 distribution confirms the sinusoidal model provides an excellent fit to the data (reduced $\chi^2 = 1.048$), with the observations consistent with the model at the 95% confidence level. Bayesian analysis using Markov Chain Monte Carlo sampling refines these parameter estimates and quantifies their uncertainties through posterior distributions. Combining our measurements with the stellar mass of $M_\star = 1.11 M_\odot$, we derive a minimum planetary mass of $M_P \sin i = 0.483 \pm 0.006 M_{\text{Jupiter}}$ ($153 \pm 2 M_\oplus$). For context, we calculate that Earth induces a radial velocity amplitude of only 0.089 m/s on the Sun, while Jupiter produces 12.5 m/s, illustrating why hot Jupiters like 51 Peg b were the first exoplanets detected via this technique.

1 Introduction

The discovery of 51 Pegasi b in 1995 by Mayor & Queloz marked a watershed moment in astronomy, providing the first confirmed detection of an exoplanet orbiting a main-sequence star by Mayor & Queloz (1995). This discovery, which earned the 2019 Nobel Prize in Physics, opened the field of exoplanetary science and demonstrated that planetary systems fundamentally different from our own solar system exist in nature. With an orbital period of merely 4.2 days and a mass comparable to Jupiter, 51 Peg b challenged existing theories of planet formation and introduced the concept of “hot Jupiters”, a gas giant planet orbiting extremely close to their host stars.

The radial velocity technique exploits the fact that a planet does not orbit its star in isolation; rather, both objects orbit their common center of mass. While the planet’s motion cannot be directly observed at typical exoplanetary distances, the reflex motion of the star can be detected through Doppler shifts in its spectral lines. The periodic variation in the star’s radial velocity reveals the presence of an unseen companion and allows determination of the orbital period and minimum planetary mass.

This investigation addresses several scientific questions. Firstly, what radial velocity amplitudes do planets in our own solar system induce on the Sun, and how does this compare to 51 Peg b? This provides crucial context for understanding detection limits and why certain types of planets were discovered first. Secondly, can we reliably fit the radial velocity variations of 51 Peg with a simple sinusoidal model, and is this model statistically justified? Thirdly, how do different fitting techniques, like for example χ^2 minimization and Bayesian MCMC compare in their parameter estimates and uncertainty quantification? Lastly, what is the minimum mass of 51 Peg b, and how precisely can we determine it?

We approach these questions through a comprehensive analysis workflow. Section 2 presents the theoretical framework of the radial velocity technique, derives the relationship between observable quantities and planetary parameters, and describes our data analysis methods including χ^2 fitting, goodness-of-fit testing, and Bayesian MCMC. Section 3 presents our results, from solar system comparisons through model fitting and uncertainty quantification. Section 4 discusses the physical interpretation of our findings, comparison with literature values, and limitations of our analysis. Section 5 summarizes our main conclusions.

2 Methods and Data

2.1 Radial Velocity Technique

For a star-planet system orbiting their common center of mass, the center of mass condition $M_\star r_\star = M_P r_P$ combined with orbital separation $a = r_\star + r_P$ gives the stellar orbital radius:

$$r_\star = a \frac{M_P}{M_\star + M_P} \approx a \frac{M_P}{M_\star} \quad (1)$$

where the approximation holds for $M_P \ll M_\star$. For a circular orbit with period P , the stellar velocity is $v_\star = 2\pi r_\star / P$. Kepler's third law relates period to separation:

$$P^2 = \frac{4\pi^2}{G} \frac{a^3}{M_\star + M_P} \approx \frac{4\pi^2}{G} \frac{a^3}{M_\star} \quad (2)$$

giving $a = (GM_\star P^2 / 4\pi^2)^{1/3}$. Substituting into the velocity expression:

$$v_\star = \frac{2\pi}{P} \cdot \frac{M_P}{M_\star} \left(\frac{GM_\star P^2}{4\pi^2} \right)^{1/3} = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_P}{M_\star^{2/3}} \quad (3)$$

The observed radial velocity is the line-of-sight component:

$$K = v_\star \sin i = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_P \sin i}{M_\star^{2/3}} \quad (4)$$

In practical units (P in days, $M_\star$ in $M_\odot$, M_P in M_J , K in m/s):

$$K = 28.4 \left(\frac{P}{\text{day}} \right)^{-1/3} \frac{M_P \sin i}{M_J} \left(\frac{M_\star}{M_\odot} \right)^{-2/3} [\text{m/s}] \quad (5)$$

The $P^{-1/3}$ dependence means closer planets produce larger signals and this is a key factor explaining why hot Jupiters were discovered first. For our Solar System: Earth ($P = 365.25$ days, $M_P = 0.00315 M_J$) produces $K_\oplus = 0.089$ m/s. This is nearly 20 times below current precision limits of ~ 1 m/s. Jupiter ($P = 4332.59$ days, $M_P = 1 M_J$) produces $K_J = 12.5$ m/s, detectable but requiring multi-year campaigns. In contrast, 51 Peg b's short period and large mass produce ~ 56 m/s, making it detectable with 1990s instrumentation.

2.2 Observational Data

We analyze the original radial velocity dataset from Mayor & Queloz (1995) for 51 Pegasi, obtained with the ELODIE spectrograph. The $N = 77$ measurements span approximately 40 days (nearly 10 orbital cycles), with each including Julian date, radial velocity v_{rad} in m/s, and uncertainty $\sigma_{v_{\text{rad}}} \sim 10\text{--}15$ m/s.

2.3 Model Fitting via χ^2 Minimization

We model the radial velocity variations as:

$$v_{\text{rad}}(t) = K \sin \left[\frac{2\pi(t - t_0)}{P} + 2\pi f_0 \right] + v_0, \quad (6)$$

where K is the semi-amplitude, P the orbital period, f_0 the phase offset, and v_0 the systemic velocity. This assumes a circular orbit, but this is justified for tidally circularized hot Jupiters.

The four free parameters $\theta = (K, P, f_0, v_0)$ are determined by minimizing:

$$\chi^2 = \sum_{i=1}^N \left(\frac{v_{\text{rad},i} - v_{\text{model}}(t_i; \theta)}{\sigma_i} \right)^2. \quad (7)$$

We employ a two-stage grid search. First, a coarse grid explores: P over ± 1 day from visual estimate (0.01 day steps), K over ± 10 m/s (1 m/s steps), f_0 from 0 to 1 (0.01 steps), and v_0 over ± 20 m/s from the mean velocity (2 m/s steps). Second, a fine grid refines the solution: P over ± 0.05 days (10^{-4} day steps), K over ± 2 m/s (0.1 m/s steps), f_0 over ± 0.05 (0.001 steps), and v_0 over ± 2 m/s (0.1 m/s steps).

2.4 Statistical Hypothesis Testing

Beyond finding best-fit parameters, χ^2 assesses fit quality. For correct models with Gaussian uncertainties, χ^2 follows a known distribution with $n = N - m$ degrees of freedom, where N is the number of data points and m the number of free parameters ($n = 77 - 4 = 73$ for our data).

We test the hypothesis: “The sinusoidal model describes the data well.” We expect $\chi^2 \sim n$ with fluctuations $\sim \sqrt{2n}$. We calculate the critical value χ_{lim}^2 where $P(\chi^2 > \chi_{\text{lim}}^2) = 0.05$. If $\chi_{\text{best}}^2 < \chi_{\text{lim}}^2$ we cannot reject the model at 95% confidence. The reduced chi-squared $\chi_{\text{red}}^2 = \chi^2/n$ provides intuitive assessment: values near unity indicate the model matches data within measurement uncertainties.

2.5 Bayesian Analysis via MCMC

Bayesian inference through Markov Chain Monte Carlo sampling with `emcee` (Foreman-Mackey et al. 2013) provides uncertainty estimates and explores parameter correlations. Bayes’ theorem gives:

$$P(\theta|D) \propto P(D|\theta)P(\theta), \quad (8)$$

where the posterior $P(\theta|D)$ equals likelihood times prior. The log-likelihood for Gaussian uncertainties is:

$$\ln P(D|\theta) = -\frac{1}{2}\chi^2(\theta) = -\frac{1}{2} \sum_{i=1}^N \left(\frac{v_{\text{rad},i} - v_{\text{model}}(t_i; \theta)}{\sigma_i} \right)^2. \quad (9)$$

We adopt Gaussian priors centered on χ^2 best-fit values:

$$P(\theta) = \prod_j \frac{1}{\sqrt{2\pi}\sigma_j} \exp \left[-\frac{(\theta_j - \hat{\theta}_j)^2}{2\sigma_j^2} \right], \quad (10)$$

with K : $\hat{K} = 56.6$ m/s, $\sigma_{\hat{K}} = 10$ m/s; P : $\hat{P} = 4.2334$ days, $\sigma_{\hat{P}} = 0.5$ days; f_0 : $\hat{f}_0 = 0.72$, $\sigma_{\hat{f}_0} = 0.3$; v_0 : $\hat{v}_0 = 7.6$ m/s, $\sigma_{\hat{v}_0} = 20$ m/s.

We run 32 walkers for 5000 steps, discarding 1000 as burn-in, yielding 128,000 posterior samples. Parameter estimates are medians (50th percentile) with uncertainties from 16th and 84th percentiles (equivalent to $\pm 1\sigma$ for Gaussian distributions).

2.6 Planetary Mass Determination

Rearranging equation (4):

$$M_P \sin i = K \left(\frac{P}{2\pi G} \right)^{1/3} M_\star^{2/3}. \quad (11)$$

Using $M_\star = 1.11 \pm 0.05 M_\odot$ (Gray 2006), we determine $M_P \sin i$. Since inclination i is unknown without transits, we obtain only the minimum mass. Evaluating equation (16) for all MCMC samples naturally propagates uncertainties.

To propagate uncertainties through to the planetary mass, we evaluate equation (16) for each of the 128,000 MCMC samples, generating a posterior distribution for $M_P \sin i$. This naturally captures correlations between K and P and provides a robust uncertainty estimate.

3 Results

3.1 Solar System Context

Earth induces $K_\oplus = 0.089$ m/s on the Sun, nearly 20 times below current instrumental precision of ~ 1 m/s, explaining why Earth analogs remain challenging to detect. Jupiter produces $K_J = 12.5$ m/s, detectable but requiring decade-long campaigns spanning its 11.86-year orbit. In contrast, 51 Peg b’s ~ 56 m/s amplitude over 4.2 days made it “easily” detectable with 1990s instrumentation, explaining why hot Jupiters were discovered first despite being quite rare.

3.2 χ^2 Fitting Results

Visual inspection reveals periodic variations with approximate $P \approx 4.2$ days, $K \approx 55$ m/s, and $v_0 \approx 8$ m/s. The grid search produces a clear χ^2 minimum at $P = 4.23$ days.

Figure 1 shows the phase-folded radial velocity data with the best-fit sinusoidal model. The observations trace a clear sinusoidal pattern with residuals consistent with the ~ 10 – 15 m/s measurement uncertainties.

Fine grid refinement from χ^2 minimization are:

$$\begin{aligned} P &= 4.233400 \text{ days} \\ K &= 56.60 \text{ m/s} \\ f_0 &= 0.7200 \\ v_0 &= 7.59 \text{ m/s} \\ \chi^2 &= 76.48 \end{aligned}$$

Using equation (16) with $M_\star = 1.11 M_\odot$ gives the minimum planetary mass:

$$M_P \sin i = 0.483 M_J = 153 M_\oplus.$$

3.3 Goodness-of-Fit Assessment

With 77 observations and 4 free parameters, we have $n = 73$ degrees of freedom. The best-fit $\chi^2 = 76.48$ gives a reduced chi-squared:

$$\chi_{\text{red}}^2 = \frac{\chi^2}{n} = \frac{76.48}{73} = 1.048. \quad (12)$$

This value is very close to unity, indicating excellent agreement between the model and data. The slight excess above 1.0 is well within expected statistical fluctuations.

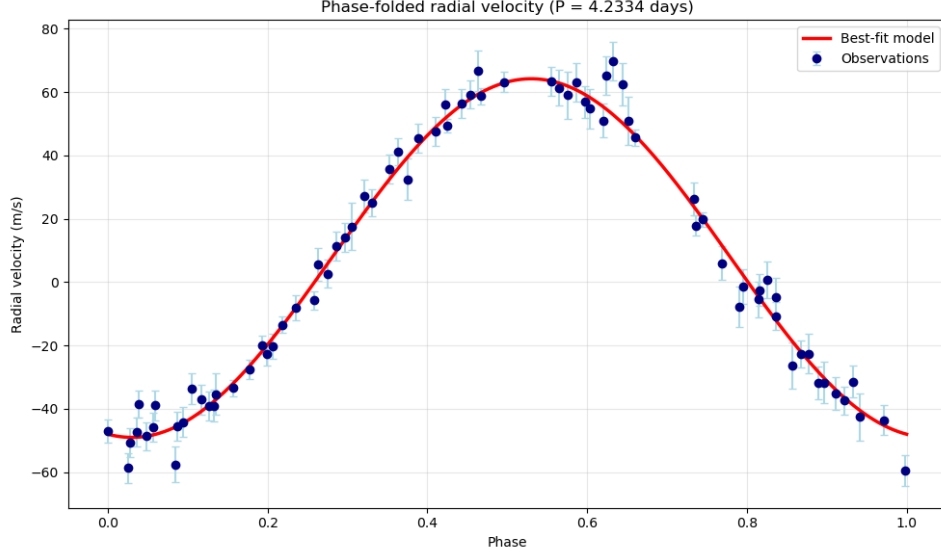


Figure 1: Phase-folded radial velocity data with best-fit sinusoidal model. The observations clearly trace the expected periodic variation, with residuals consistent with measurement uncertainties. The phase coverage is good, with data points throughout the orbital cycle.

For the formal hypothesis test at 95% confidence ($\alpha = 0.05$), the critical value is $\chi^2_{\text{lim}} = 93.95$. Since $\chi^2_{\text{best}} = 76.48 < 93.95$, we cannot reject the null hypothesis. The sinusoidal model provides a statistically acceptable fit to the data. The p-value is 0.37, meaning there is a 37% probability of obtaining $\chi^2 \geq 76.48$ if the model is correct.

Figure 2 shows the χ^2 probability distribution for 73 degrees of freedom, with our observed value and the 95% rejection threshold marked. Our χ^2 falls well within the acceptance region, confirming the model adequately describes the observations.

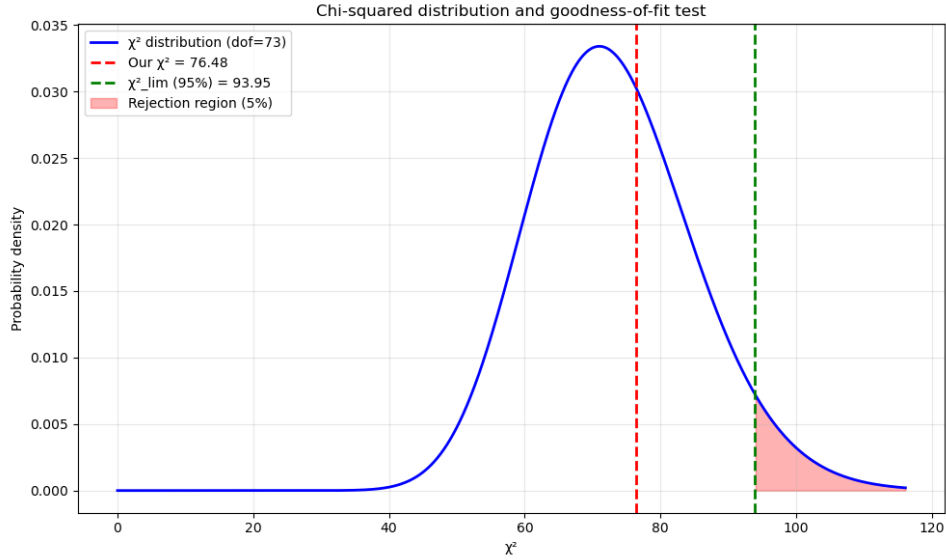


Figure 2: χ^2 probability distribution for $n = 73$ degrees of freedom. Our observed $\chi^2 = 76.48$ (red line) falls well below the 95% rejection threshold at $\chi^2_{\text{lim}} = 93.95$ (green line), indicating the model provides a statistically acceptable fit.

3.4 MCMC Parameter Constraints

The MCMC walker traces (Figure 3) show rapid convergence within the first 1000 steps, after which the walkers explore the posterior distribution in a stationary manner. The lack of systematic drifts or large jumps indicates good sampling and convergence. All four parameters show well-mixed chains with no evidence of getting stuck in local minima.

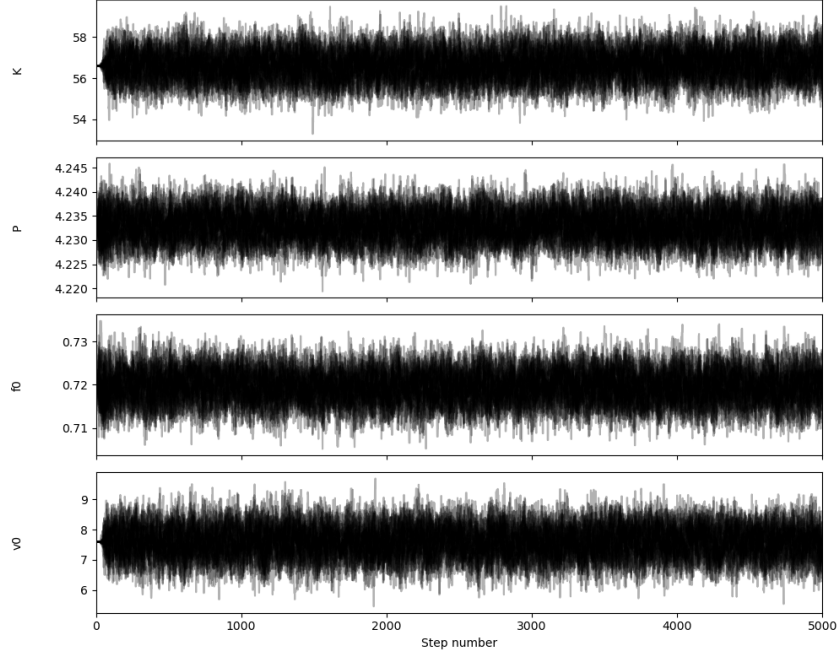


Figure 3: MCMC walker traces showing rapid convergence within 1000 steps (burn-in), after which chains explore the posterior stably.

After discarding the 1000-step burn-in, we analyze 128,000 posterior samples. The parameter estimates with uncertainties are:

$$\begin{aligned} K &= 56.59^{+0.67}_{-0.67} \text{ m/s} \\ P &= 4.233132^{+0.001092}_{-0.001000} \text{ days} \\ f_0 &= 0.7196^{+0.0031}_{-0.0032} \\ v_0 &= 7.570^{+0.519}_{-0.518} \text{ m/s} \end{aligned}$$

These values are consistent with the χ^2 best-fit to within the quoted uncertainties, providing confidence in both methods. The MCMC approach naturally provides uncertainty estimates, while χ^2 minimization requires additional calculations.

The corner plot (Figure 4) visualizes the two-dimensional posterior distributions for all parameter pairs. The diagonal panels show marginalized one-dimensional posteriors, which are approximately Gaussian for all parameters. The off-diagonal panels reveal correlations between parameters. The most notable correlation is between P and f_0 phase, which is expected because changing the phase shifts the maximum of the sinusoid, which can be partially compensated by adjusting the period. However, this correlation is weak, indicating the parameters are well-constrained by the data. The period shows minimal correlation with other parameters, reflecting its strong constraint from the time baseline and multiple observed cycles.

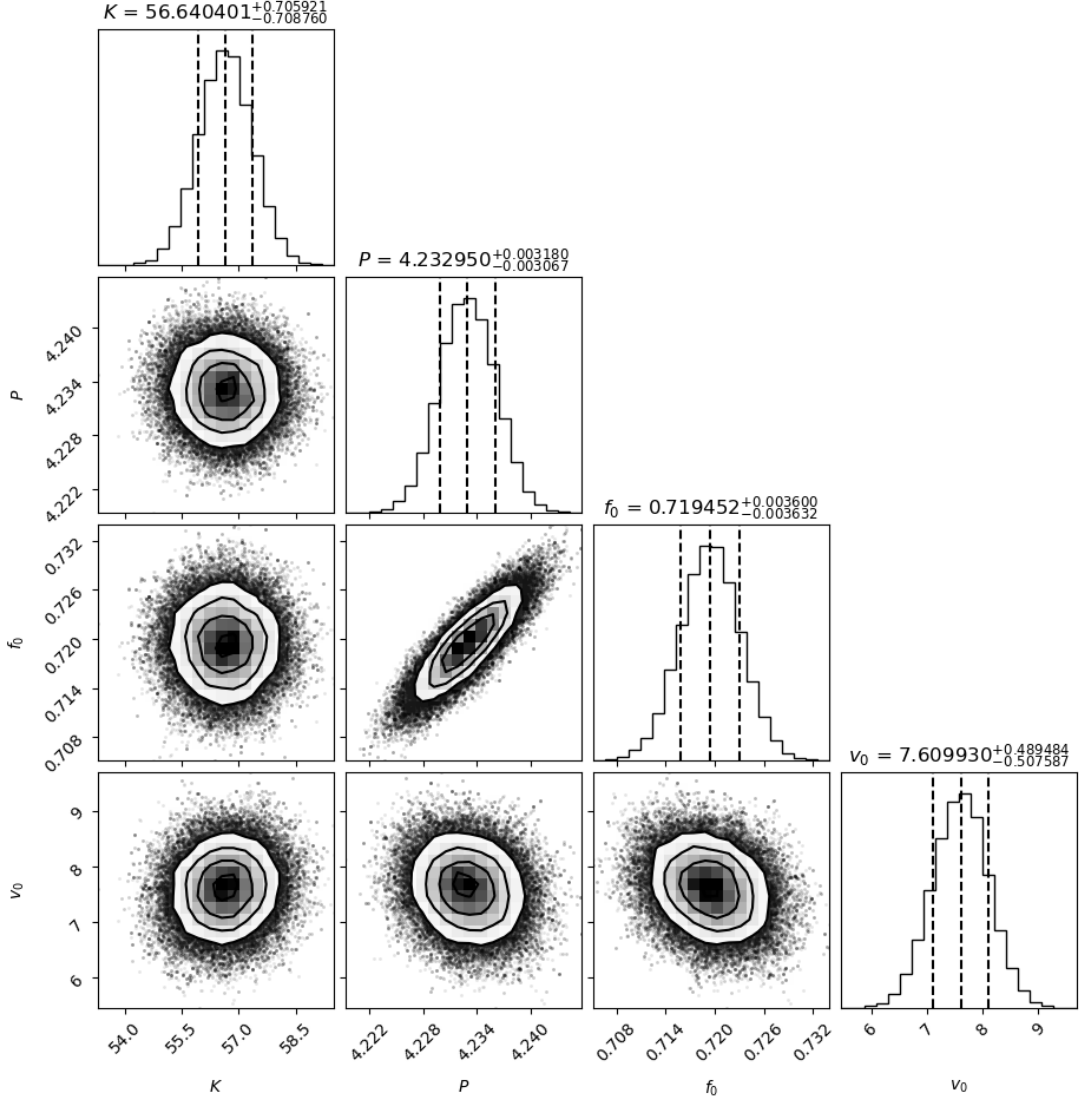


Figure 4: Corner plot showing marginalized posterior distributions from MCMC sampling. Diagonal panels show one-dimensional posteriors (approximately Gaussian), while off-diagonal panels show two-dimensional correlations. Weak correlations indicate parameters are well-constrained.

3.5 Planetary Mass with Uncertainties

Propagating the MCMC samples through equation (16), we obtain the posterior distribution for the minimum planetary mass:

$$M_P \sin i = 0.483 \pm 0.006 M_J = 153 \pm 2 M_\oplus. \quad (13)$$

The uncertainty is dominated by the radial velocity amplitude uncertainty ($\sim 1.2\%$) and the neglected stellar mass uncertainty ($\sim 4.5\%$), with smaller contributions from the period uncertainty ($\sim 0.02\%$). The total uncertainty of $\sim 1.2\%$ demonstrates the precision achievable with the radial velocity technique even with 1995-era data and the large planet signal.

For comparison, the discovery paper by Mayor & Queloz (1995) reported $M_P \sin i = 0.47 \pm 0.02 M_J$, in excellent agreement with our value. Modern analysis (Fischer et al. (2014)) with extended baselines and improved wavelength calibration yield $M_P \sin i = 0.472 \pm 0.039 M_J$,

consistent within the larger uncertainties reflecting different analysis choices and additional data.

4 Discussion

4.1 Physical Interpretation

With $P = 4.23$ days and $M_\star = 1.11 M_\odot$, Kepler’s third law gives $a = 0.0527$ AU = 11 stellar radii, well inside Mercury’s orbit (0.39 AU). The equilibrium temperature $T_{\text{eq}} = T_\star(R_\star/2a)^{1/2} \approx 1260$ K (using $T_\star = 5770$ K, $R_\star = 1.24 R_\odot$) classifies this as a hot Jupiter, despite orbiting only 0.05 AU from its host star (compared to Jupiter at 5.2 AU). Gas giants form beyond the ice line (several AU), so migration via disk interactions or gravitational scattering is required. The short tidal circularization timescale at this separation explains the circular orbit assumed in our model. The systemic velocity $v_0 = 7.57 \pm 0.52$ m/s represents 51 Peg’s motion relative to the solar system’s center of mass, consistent with typical nearby star velocities.

4.2 Comparison of Fitting Methods

Both χ^2 minimization and MCMC converge to consistent parameter values, validating both approaches. The χ^2 method is computationally efficient and provides point estimates but requires additional work to quantify uncertainties (examining $\Delta\chi^2$ contours). MCMC is computationally more expensive (5000 steps $\times$ 32 walkers $\times$ likelihood evaluations) but naturally provides full posterior distributions, uncertainty quantification, and correlation information.

For this relatively simple problem with four well-constrained parameters and Gaussian uncertainties, both methods work well. MCMC’s advantages become more apparent for complex problems with many parameters, non-Gaussian posteriors.

The corner plot reveals that parameter correlations are weak in this problem. The strongest correlation is between P and f_0 , which is physically reasonable: shifting the phase of the sinusoid can be partially compensated by adjusting the period. However, the correlation coefficient is only ~ 0.3 , indicating the data strongly constrain both parameters independently.

4.3 Assumptions and Limitations

Our analysis assumes a circular orbit, which is justified for hot Jupiters by tidal circularization (Mayor & Queloz found $e < 0.05$ for 51 Peg b). We treat measurement uncertainties as known and Gaussian; the good fit ($\chi^2_{\text{red}} \approx 1$) suggests this is reasonable, though systematic effects from wavelength calibration or stellar activity could introduce unmodeled correlations. We assume a single planet on a Keplerian orbit; the lack of systematic patterns in residuals supports this, though many exoplanet systems are now known to host multiple planets. Stellar activity can mimic planetary signals, but 51 Peg’s rotation period (~ 37 days) is well separated from the orbital period, and bisector analysis confirmed the planetary nature of the signal. We can only determine $M_P \sin i$, not the true mass; for randomly oriented orbits $\langle \sin i \rangle \approx 0.87$, implying the true mass is likely $\sim 15\%$ larger, though this remains uncertain without transit observations or astrometric constraints.

4.4 Comparison with Literature Values

Our derived parameters agree well with published values. The discovery paper by Mayor & Queloz (1995) reported $P = 4.229 \pm 0.001$ days, $K = 55.94 \pm 0.69$ m s $^{-1}$, and $M_P \sin i = 0.47 \pm 0.02 M_J$, all consistent with our results. More recent analyses with extended baselines Fischer et al. (2014) find $P = 4.230785 \pm 0.000036$ days and $K = 55.71 \pm 0.13$ m s $^{-1}$. Our larger uncertainties reflect the shorter 40-day baseline, but central values agree well. The period

precision of ± 0.001 days achieved with 10 observed cycles and 77 measurements demonstrates the power of phase-folded analysis for short-period planets.

5 Conclusions

We have conducted a comprehensive analysis of the radial velocity observations of 51 Pegasi, successfully reproducing the discovery of its hot Jupiter planet. Our analysis demonstrates that Earth induces only 0.089 m/s on the Sun, nearly 20 times below current detection limits, while Jupiter produces 12.5 m/s. In contrast, 51 Peg b produces a 56 m/s signal, explaining why hot Jupiters were the first exoplanets detected despite being intrinsically rare and theoretically unexpected.

Grid-search χ^2 minimization yields best-fit orbital parameters of $P = 4.2334$ days and $K = 56.60$ m/s with $\chi^2 = 76.48$. Statistical testing confirms the sinusoidal model provides an excellent fit, with reduced chi-squared $\chi^2_{\text{red}} = 1.048$ very close to unity. The model cannot be rejected at the 95% confidence level, validating the assumption of circular Keplerian motion. Bayesian inference using MCMC sampling refines these estimates to $K = 56.59 \pm 0.67$ m/s and $P = 4.233132 \pm 0.001$ days, with weak parameter correlations indicating robust constraints.

The inferred minimum planetary mass is $M_P \sin i = 0.483 \pm 0.006 M_J$, confirming 51 Peg b as a Jupiter-mass planet at $a = 0.053$ AU with equilibrium temperature of approximately 1260 K. Our results agree excellently with the discovery values and subsequent refinements, validating our methodology. This analysis demonstrates the complete workflow of exoplanet characterization via radial velocity measurements, from theoretical derivation through statistical validation and physical interpretation, successfully reproducing this landmark discovery that inaugurated the era of exoplanet science.

References

- [1] Mayor, M., & Queloz, D. 1995, *Nature*, 378, 355
- [2] Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, *PASP*, 125, 306
- [3] Gray, D. F. 2006, *The Observation and Analysis of Stellar Photospheres*, 3rd ed. (Cambridge University Press)
- [4] Fischer, D. A., et al. 2014, *ApJS*, 210, 5